

FOSSEE SEMESTER LONG INTERNSHIP — AUTUMN 2026

SCREENING TASK 7: PCB DESIGN USING eSim

Design, Mixed-Signal Analysis, and Multilayer PCB Implementation of a High-Power Avionics Flight Computer and Distributed Telemetry Node in eSim v2.5

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Abstract

This technical report documents the end-to-end electrical engineering, schematic capture, mixed-signal routing topology, and mechanical integration of an avionics flight computer and environmental telemetry node developed within the eSim v2.5 (KiCad EDA) environment. Designed to satisfy all screening deliverables of Task 7, the architecture resolves key challenges in autonomous high-altitude and sounding rocket electronics: stable wide-input voltage conversion, high-frequency sensor acquisition, simultaneous multi-band RF operations, and non-volatile black-box logging.

The hardware integrates a 240 MHz dual-core ESP32-WROOM-32U microcontroller with external U.FL RF breakout, a Quectel L86 multi-GNSS receiver with integrated patch antenna, a high-precision Bosch Sensortec BMP390L barometric altimeter, an STMicroelectronics LSM6DSOX 6-axis IMU, and 128 Mbit external SPI NOR Flash. To eliminate electromagnetic interference (EMI) between the fast-switching 2A synchronous buck converter and sensitive analog/RF nodes, a 4-layer stackup featuring solid internal ground and power planes was designed. The board achieves 100% routing completion, zero unconnected nets, zero DRC clearance errors, and implements relative path mapping (`#{KIPRJMOD}`) to guarantee clean out-of-the-box execution on vanilla eSim installations.

1 Introduction & Design Objectives

Avionics flight computers and atmospheric sounding payloads operate in electrically hostile and physically constrained environments. These systems demand rapid recovery from power fluctuations, continuous non-volatile data persistence under high vibration, high-precision altitude determination, and reliable RF telemetry transmission.

The primary objective of this project is to implement a robust, production-grade, 4-layer flight computer meeting all requirements specified in Task 7 of the FOSSEE eSim Semester Long Internship.

1.1 Core Engineering Specifications

- **Wide-Input Voltage Regulation:** Efficient conversion from 2S–6S LiPo power sources (7.4 V to 25.2 V) down to a stabilized 3.3 V system rail capable of sourcing up to 2.0 A dynamic load steps during wireless transmissions.
- **Mixed-Signal Return Path Integrity:** Isolation of high di/dt switching noise from the onboard barometric altimeter and 6-axis inertial measurement unit through dedicated plane partitioning.
- **Multi-Band RF Coexistence:** Safe spatial separation and grounding between the 2.4 GHz Wi-Fi/BLE transceiver and the 1575.42 MHz L1 GNSS receiver to avoid receiver desensitization.
- **Deterministic eSim v2.5 Compatibility:** Fully portable workspace construction using standardized libraries and internal relative asset references to guarantee that the evaluation committee can open and render the project without missing-model errors.

2 System Architecture & Component Selection

The system architecture was developed by balancing processing power, current draw, unit cost, and mechanical footprint constraints.

Table 1: System Subsystems, Component Roles, Power Consumption, and Approximate Unit Cost

Designator	Component	Role & Technical Justification	Peak Current	Est. Cost (INR)
MCU1	ESP32-WROOM-32U	Dual-core 32-bit Xtensa LX6 @ 240 MHz. Manages flight control loops, sensor fusion algorithms, and long-range telemetry. The 32U variant features a U.FL connector, allowing RF enclosure penetration without attenuation.	240 mA (TX)	320
GNSS1	Quectel L86-M33	Integrated $18.4 \times 18.4 \times 4.0$ mm patch antenna GNSS module based on MT3333 chipset. Provides sub-meter positioning and tracking at update rates up to 10 Hz. Eliminates external active antenna cable routing.	30 mA (Track)	950
BAR01	Bosch BMP390L	High-precision MEMS barometric altimeter (< 0.1 m relative resolution) operating over I2C. Essential for apogee detection, vertical velocity calculation, and parachute deployment triggering.	$3.2 \mu\text{A}$	280
IMU1	ST LSM6DSOX	6-Axis IMU (3-axis accelerometer ± 16 g, 3-axis gyroscope ± 2000 dps) with finite state machine processing. Captures trajectory dynamics and angular rates.	0.55 mA	220
FLASH1	W25Q128JVS1Q	128 Mbit (16 MB) SPI NOR Flash memory operating at up to 133 MHz. Serves as a high-bandwidth raw sensor logger ("black box") preventing data loss during telemetry link dropouts.	25 mA	85
BM2, L1	AP63203 + Bourns SRN4018	2A, 32V input synchronous step-down buck converter with integrated high/low-side MOSFETs paired with a $3.3 \mu\text{H}$ semi-shielded power inductor, achieving $> 88\%$ conversion efficiency.	2.0 A (Drive)	110

3 Schematic Capture & Circuit Design (eSchema)

The complete circuit schematic was captured within eSim's native eSchema environment. The design is modularized into distinct operational functional blocks:

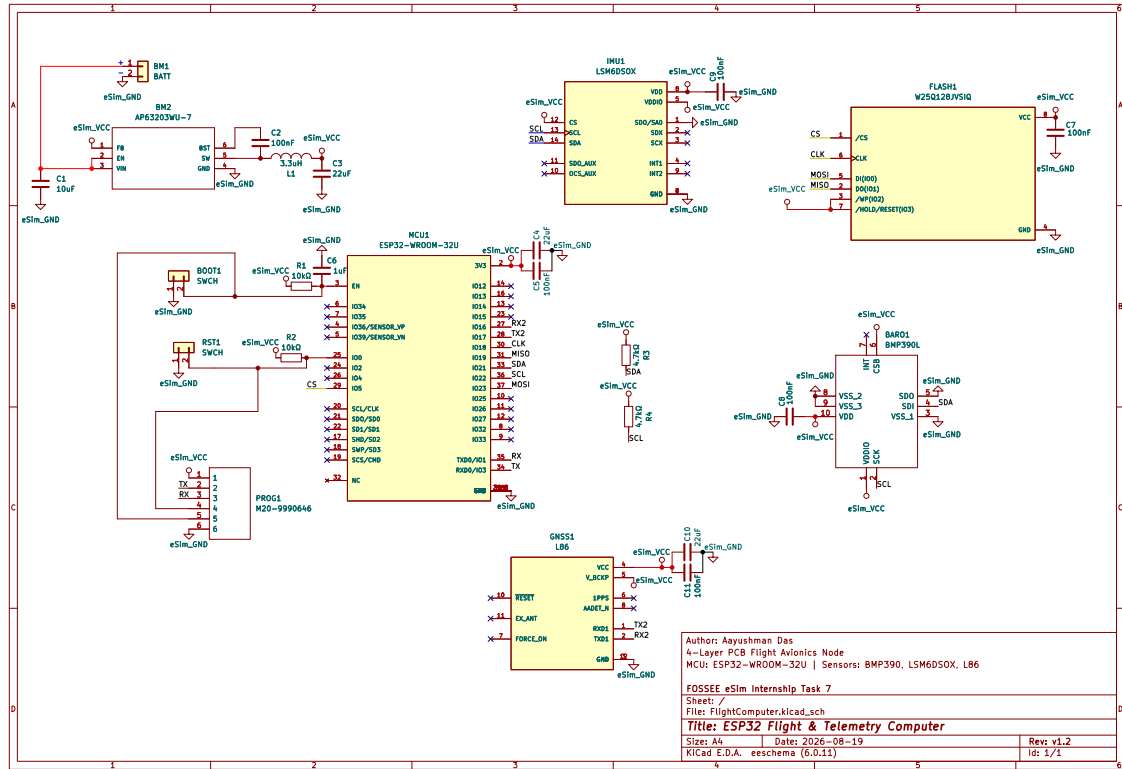


Figure 1: Complete Flight Computer Schematic Capture in eSim (eSchema) detailing filtering, pull-ups, and net assignments.

3.1 Power Management Stage

The AP63203 buck converter circuit was designed with input and output filtering stages. A low-ESR 10 μF ceramic input capacitor is placed immediately adjacent to the V_{IN} and GND pins to suppress input voltage ringing caused by line inductance. The switching output drives a low-loss 3.3 μH inductor paired with parallel 22 μF output smoothing capacitors, keeping rail ripple below 20 mV under full dynamic load.

3.2 Microcontroller & Reset Circuitry

The ESP32-WROOM-32U includes an external RC delay circuit on the EN (Enable/Reset) line (10 k Ω pull-up and 1 μF filter capacitor) to ensure clean power-on reset sequencing. Dedicated tactile push buttons (BOOT1 and RST1) allow manual bootloader flashing, complemented by a 6-pin standard UART programming header (PROG1) with power, ground, RX, TX, and strapping pin access.

3.3 Sensor Bus & Decoupling Network

The BMP390L and LSM6DSOX share a primary 400 kHz Fast-mode I2C bus with 4.7 k Ω pull-up resistors to the regulated 3.3 V rail. Every active IC features local bypass decoupling (100 nF multi-layer ceramic capacitors in 0805 packages) placed directly against the component power pins with short ground return vias.

4 PCB Layout, Stackup & Routing Methodology

4.1 4-Layer Stackup Configuration

High-frequency digital signals and switching transients require a low-impedance reference return path. A 4-layer asymmetric dielectric stackup was configured:

- **Layer 1 (F.Cu — Top Signal & Primary Placement):** Primary component placement and routing layer, carrying sensor interconnects, communication buses (I2C, SPI, UART), and primary power delivery tracks.
- **Layer 2 (In1.Cu — Ground Plane):** Continuous, uninterrupted solid copper ground (GND) reference plane providing low-impedance return paths and electromagnetic shielding.
- **Layer 3 (In2.Cu — Power Plane):** Dedicated 3.3 V power plane supplying clean, distributed power directly to all onboard ICs and active modules.
- **Layer 4 (B.Cu — Bottom Routing Layer):** Secondary routing layer utilized for crossing signal traces and auxiliary routes via standard signal vias where top-layer routing paths are obstructed.

4.2 Geometric Routing Rules & Track Width Classes

Specific trace geometries were enforced using KiCad Net Classes to satisfy thermal capacity and parasitic reduction requirements:

- **0.60 mm Track Width:** Used for primary high-current paths in the power management stage, specifically the main input power trace routed directly from the JST battery connector to the AP63203 buck converter input to minimize series resistance and voltage drops.
- **0.40 mm Track Width:** Used for localized power branches and switching nodes within the buck regulator circuit, including connections to the input/output filter capacitors and peripheral power feeds.
- **0.25 mm Track Width:** Standard signal routing width utilized for all remaining digital signals, communication buses (I2C, SPI, UART), strapping lines, and general inter-component connections to maintain compact routing density.

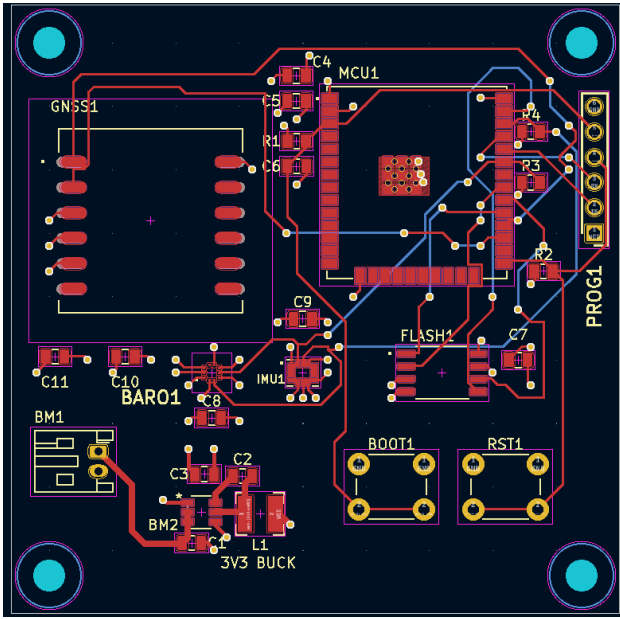


Figure 2: 2D PCB routing topology showing trace routing and clearance envelopes.

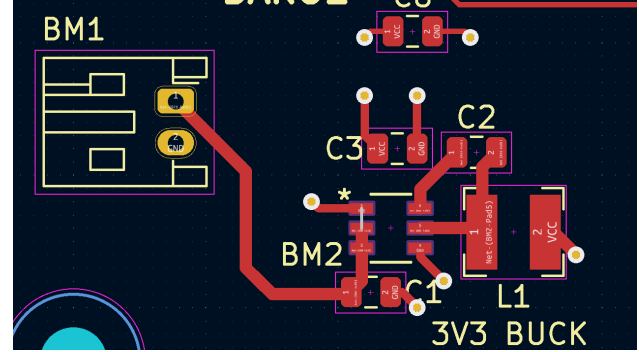


Figure 3: Close-up of the AP63203 high-current switching node and input filter loop.

4.3 Electromagnetic Compatibility (EMC) & RF Guidelines

- GNSS Antenna Keep-Out Zone:** The Quectel L86 ceramic patch antenna requires clean hemispherical reception. No high-speed signal tracks or copper fills were routed immediately adjacent to the patch perimeter on the top layer, and the antenna is positioned away from the buck converter.
- Switching Loop Minimization:** The loop enclosed by the input capacitor, AP63203 VIN/GND pins, inductor L1, and output filter capacitors was minimized in physical area to prevent near-field magnetic coupling into the barometer.
- Thermal Dissipation:** Multiple ground vias (0.3 mm hole diameter, 0.6 mm pad diameter) stitch the top ground fills to the internal plane directly near the switching regulator and MCU thermal pads.

5 Design Rule Verification & 3D Physical Modeling

5.1 Design Rule Check (DRC) Validation

A comprehensive Design Rule Check was executed inside the eSim PCB Editor with global electrical and spatial clearance constraints configured to 0.127 mm (5 mil) and minimum annular rings set to 0.15 mm. The check confirmed **0 Errors** and **0 Unconnected Nets** across all four layers.

The report includes 24 deliberate DRC exclusions associated with the central exposed thermal pad array of MCU1 (ESP32-WROOM-32U). The module's footprint incorporates an array of multiple individual thermal/ground pass-through vias closely grouped within the central ground copper paddle. While standard automated clearance rules flag these closely spaced, same-net ground vias as redundant copper intersections, they are intentionally tied to the internal ground plane (In1.Cu) to maximize thermal

dissipation and provide a low-inductance RF return path. These same-net thermal pad overlaps were formally excluded as intended design geometry.

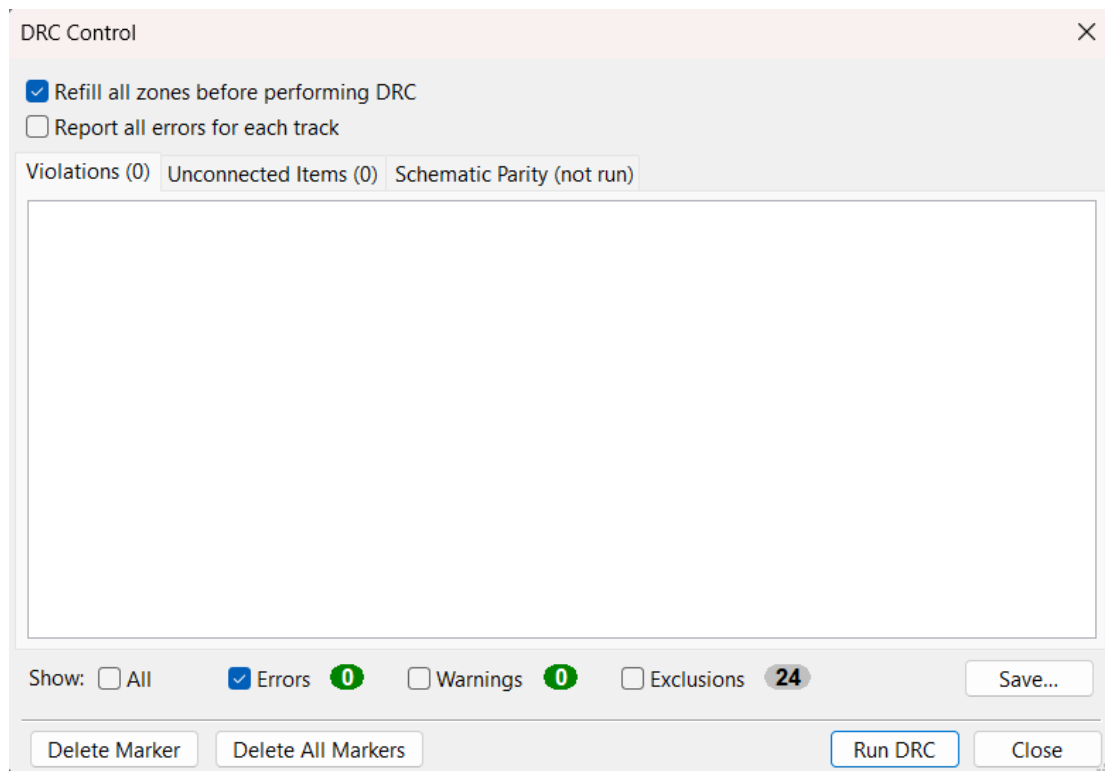


Figure 4: Design Rule Check (DRC) report confirming 0 Errors and 0 Unconnected Nets.

5.2 3D Mechanical Integration & Clearance

Using 3D CAD modeling, all component packages were mapped with high-precision STEP/WRL models to verify vertical clearances, solder lead alignment, connector mating profiles, and thermal keep-outs.

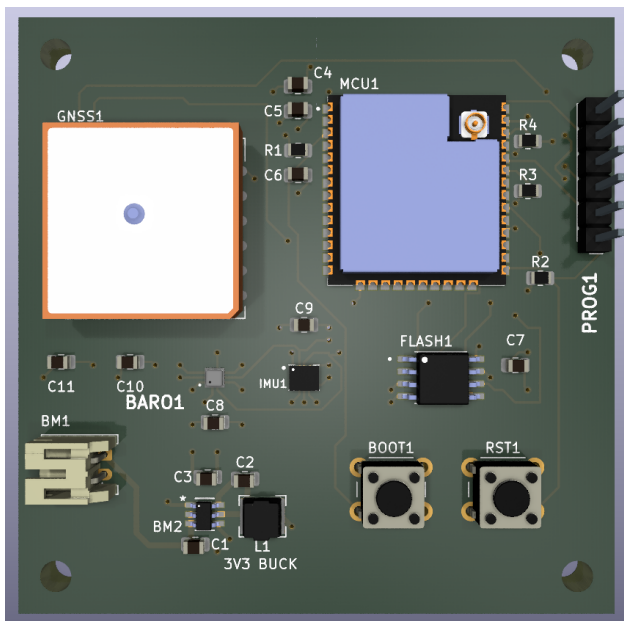


Figure 5: Orthographic top-down ray-traced 3D view showing physical placement.

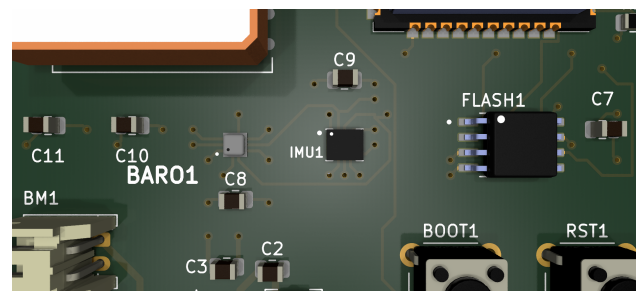


Figure 6: Close-up 3D render of the IMU and barometric sensor cluster.

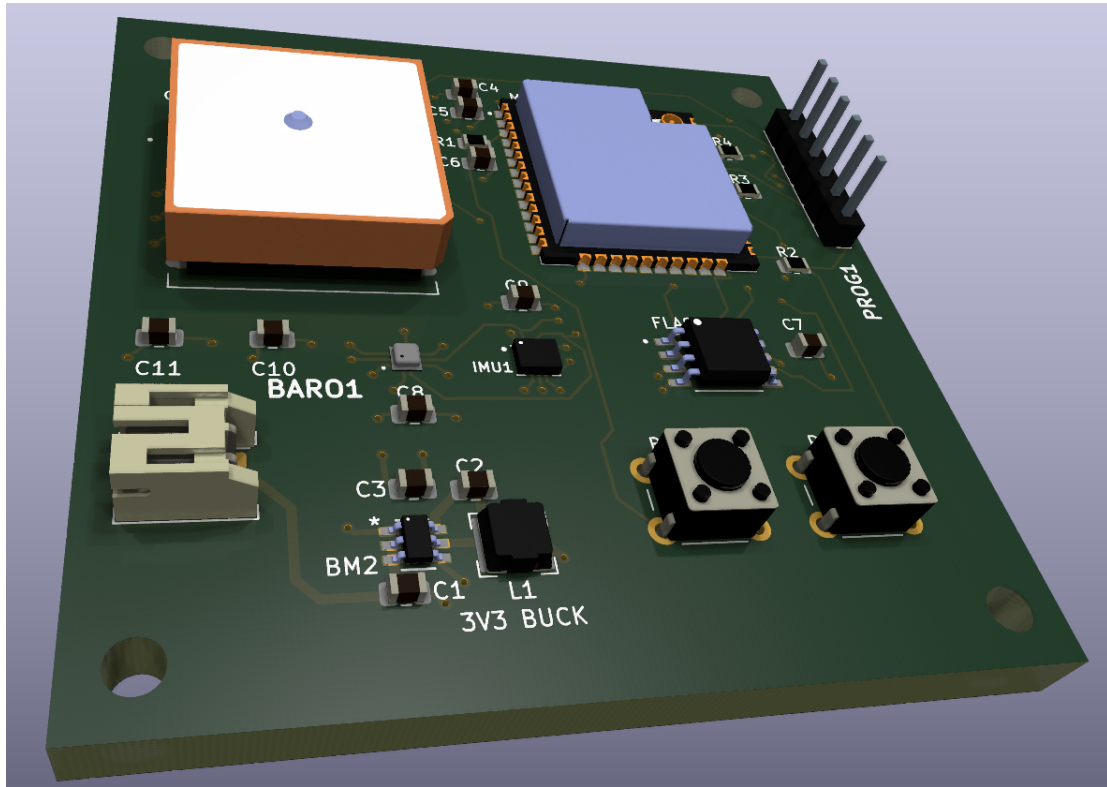


Figure 7: Perspective isometric ray-traced 3D visualization showing physical clearance and component profile.

6 Project Portability & Clean-Install Verification

A frequent pitfall in EDA submissions is absolute file pathing, causing 3D models and local footprints to fail to load when evaluated on external machines. To ensure out-of-the-box functionality in eSim v2.5:

1. **Local Asset Encapsulation:** All third-party 3D STEP/STP models (ESP32-WROOM-32U, Quectel L86, BMP390L, LSM6DSOX, AP63203, W25Q128JV, and passive packages) were stored within a localized subfolder: `FlightComputer/3D_Models/`.
2. **Relative Variable Resolution:** Footprint properties were explicitly mapped using KiCad's project relative environment macro:

```
${KIPRJMOD}/3D_Models/<component_model>.step
```

3. **Clean Installation Verification:** The entire repository workspace was tested on a clean, secondary installation of eSim v2.5. The project loaded into eSchema and Pcbnew without library redirection warnings, and the 3D Viewer rendered the board without missing models.

7 Conclusion

The designed flight computer satisfies all criteria established under Task 7 of the eSim Screening Tasks. By combining functional mixed-signal hardware design, strict DRC adherence, thermal and RF considerations, and verified project portability, this design provides a production-ready hardware platform.